

Independence of Local SAT Channels from MAAT Supports

Is the Missing MOMS Channel Independent of MAAT Supports?

MAAT Structural Selection Series – Paper 71

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Abstract

Paper 69 executed the preregistered MAAT v1.7 Gate Challenge protocol on a public SATLIB smoke subset and found no support for the frozen root-level gate as a practical CDCL improvement. Paper 70 diagnosed the negative result: MOMS appears to exploit decision-local short-clause and occurrence geometry that the root-level gate does not directly encode. The present paper asks the next, narrower question: *is that local SAT channel independent structural information, or mostly a reparameterization of the existing MAAT supports $H, B, S, V, R_{\text{rob}}$?*

This paper does not modify the preregistered gate, does not propose a new gate, and does not define a new MAAT version. It is a post-hoc hypothesis-generation study. Using only derived Paper-70 outputs, we regress candidate local channels against $(H, B, S, V, R_{\text{rob}}, G_{\text{gate}})$, compute residual correlations with the gate-to-MOMS regret, and compare support-only, local-only, and combined models under leave-one-family-out evaluation.

The result is deliberately cautious. Many local channels are strongly reconstructible from the existing support language: degree coefficient of variation has $R^2 = 0.964$ from supports, short-clause pressure has $R^2 = 0.917$, and Jeroslow–Wang concentration has $R^2 = 0.958$. The strongest Paper-70 raw channel, shortest-clause variable entropy, has no residual signal after support removal ($\rho_S = -0.014$, shuffle $p = 0.899$). Only weak residual candidates remain: variable-occurrence CV ($\rho_S = -0.264$, shuffle $p = 0.015$) and degree CV ($\rho_S = 0.248$, shuffle $p = 0.049$). These residuals are hypothesis-generating only; they are not robust evidence for a new support, a new gate, or a SAT-specific extension. Moreover, local-heavy models collapse under leave-one-family-out evaluation, showing that the apparent local signal is strongly family-dependent.

The conclusion is not that a new SAT gate has been found. Rather, Paper 71 draws a hypothesis boundary: the missing MOMS channel is partly encoded by existing supports, but small decision-local occurrence residuals may remain. Any future Gate+ L experiment must therefore be preregistered as a new test and treated as a SAT-specific hypothesis, not as a retroactive repair of Paper 69.

1 Purpose and Guardrail

This paper follows the methodological sequence:

preregistration $\rightarrow$ execution $\rightarrow$ diagnosis $\rightarrow$ hypothesis generation.

Paper 68 preregistered the Gate Challenge [4]. Paper 69 executed the first external SATLIB smoke test [5]. Paper 70 diagnosed the gate-to-MOMS gap [6].

The purpose of Paper 71 is not to improve the Paper-69 gate. It is to ask whether the missing local channel identified in Paper 70 is independent enough to justify a future preregistered hypothesis.

This paper does not modify the preregistered MAAT v1.7 gate or reinterpret the Gate Challenge outcome. It is a post-hoc diagnostic analysis intended solely to generate new, independently testable hypotheses for future preregistered experiments.

This guardrail is essential. If a new local channel is ever tested as part of a solver policy, it must be declared before execution in a new experiment.

2 Data and Target

The input is the derived Paper-70 output file

`paper70_local_channel_features.csv`

containing 30 family-balanced SATLIB smoke instances. No raw CNF files are required for Paper 71.

The target remains the Paper-69 gate-to-MOMS regret:

$$r_i = C_i(\text{gate-v1.7}) - C_i(\text{MOMS}). \quad (1)$$

Positive r_i means the frozen gate used more compute than MOMS.

The existing support basis is

$$\mathcal{S} = \{H, B, S, V, R_{\text{rob}}, G_{\text{gate}}\}. \quad (2)$$

The candidate local-channel set is

$$\mathcal{L} = \{H_{\min}, d_{\text{deg}}, d_{\text{occ}}, P_{\min}, P_{\text{MOMS}}, c_{\text{JW}}, H_{\ell}, P_{\text{w}}\}. \quad (3)$$

Operationally these correspond to shortest-clause variable entropy, degree dispersion, variable-occurrence dispersion, short-clause pressure, MOMS-local signal, Jeroslow–Wang concentration, clause-length entropy, and weighted clause pressure.

3 Independence Test

For each local channel $L \in \mathcal{L}$, we fit a support reconstruction model

$$L = f(H, B, S, V, R_{\text{rob}}, G_{\text{gate}}) + \epsilon_L, \quad (4)$$

using standardized ordinary least squares. The explanatory fraction $R^2(L|\mathcal{S})$ measures how much of the local channel is already encoded by the existing support language.

The residual

$$L_{\text{res}} = L - \hat{L}_{\mathcal{S}} \quad (5)$$

is then correlated with r_i . A local channel is treated as a candidate independent structural channel only if:

- it is not strongly reconstructed by $\mathcal{S}$, and
- its residual remains associated with gate-to-MOMS regret.

This is a hypothesis-generation criterion, not a validation criterion.

4 Results

4.1 Local channels are mostly support-reconstructible

Table 1 shows the main independence diagnostics. Several local channels are strongly predictable from existing supports. This is an important negative result: the local SAT channel should not automatically be promoted to a new primitive support.

Local channel	$\rho_S(L, r)$	$R^2(L S)$	$\rho_S(L_{\text{res}}, r)$	Interpretation
Shortest-clause variable entropy	0.643	0.625	−0.014	not independent
Degree CV	−0.567	0.964	0.248	weak residual
Variable-occurrence CV	−0.488	0.722	−0.264	weak residual
Short-clause pressure	−0.389	0.917	0.099	mostly redundant
MOMS-local signal	−0.338	0.943	0.073	mostly redundant
Jeroslow–Wang concentration	−0.217	0.958	−0.188	mostly redundant
Clause-length entropy	−0.212	0.865	0.147	mostly redundant
Weighted clause pressure	0.211	0.620	−0.050	not independent

Table 1: Local-channel independence diagnostics. The strongest raw Paper-70 channel, shortest-clause variable entropy, loses its signal after support reconstruction. Only small degree/occurrence residuals remain.

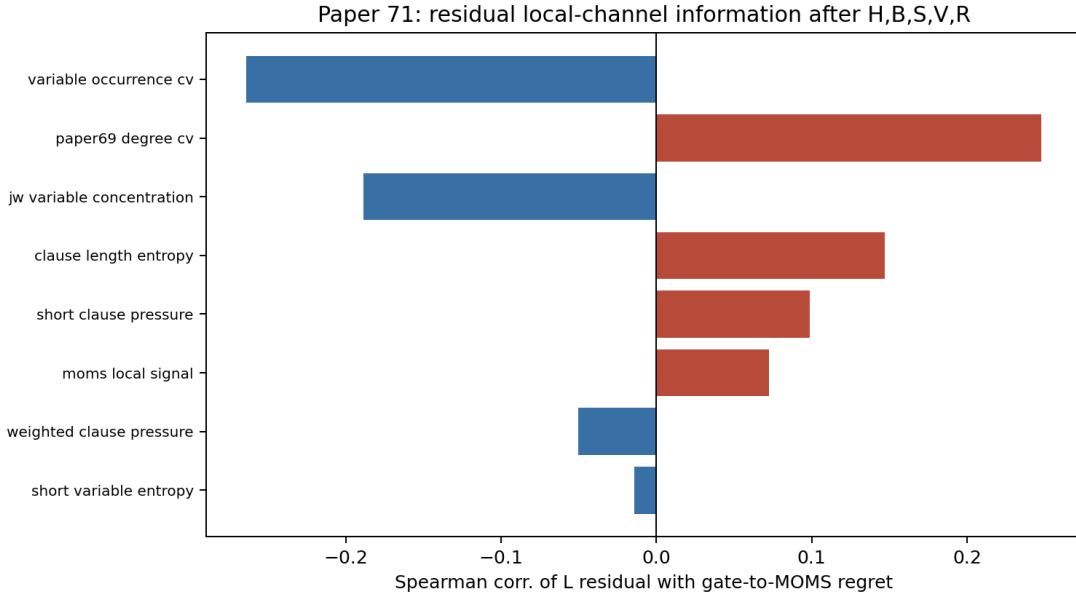


Figure 1: Residual local-channel information after removing the existing support basis ($H, B, S, V, R_{\text{rob}}, G_{\text{gate}}$). The residual signals are modest; no channel is strong enough to justify an immediate gate update.

4.2 Shuffle-null checks

The residual signals were compared against shuffled-channel nulls. The strongest Paper-70 raw channel, shortest-clause variable entropy, does not survive residual testing:

$$\rho_S = -0.014, \quad p_{\text{shuffle}} = 0.899.$$

Variable-occurrence CV and degree CV are the only weak residual candidates:

$$\begin{aligned}\rho_S^{(\text{occ})} &= -0.264, & p_{\text{shuffle}} &= 0.015, \\ \rho_S^{(\text{deg})} &= 0.248, & p_{\text{shuffle}} &= 0.049.\end{aligned}$$

These values are suggestive but not decisive. They are small-sample, post-hoc, family-structured diagnostics.

Residual channel	ρ_S	Shuffle p	95% null interval
Shortest-clause variable entropy	-0.014	0.899	$[-0.205, 0.201]$
Degree CV	0.248	0.049	$[-0.250, 0.234]$
Variable-occurrence CV	-0.264	0.015	$[-0.208, 0.218]$

Table 2: Residual shuffle-null checks. Only occurrence/degree residuals remain weakly non-null in this smoke subset.

4.3 Model comparison and family generalisation

We compare four descriptive models of gate-to-MOMS regret:

- supports only,
- supports plus gate,
- local channels only,
- supports plus local channels.

In-sample explanatory power increases when local channels are added. However, leave-one-family-out performance collapses for local-heavy models:

$$R_{\text{LOFO}}^2(\text{local only}) = -108.2, \quad R_{\text{LOFO}}^2(\text{supports} + \text{local}) = -155.7.$$

This shows that the local channels are highly family-dependent in the present smoke subset. It is the strongest negative result of the paper: local channels can improve descriptive in-sample fit while failing to transfer across SAT families. Therefore they should not be interpreted as robust structural evidence until a future preregistered experiment shows family-stable utility.

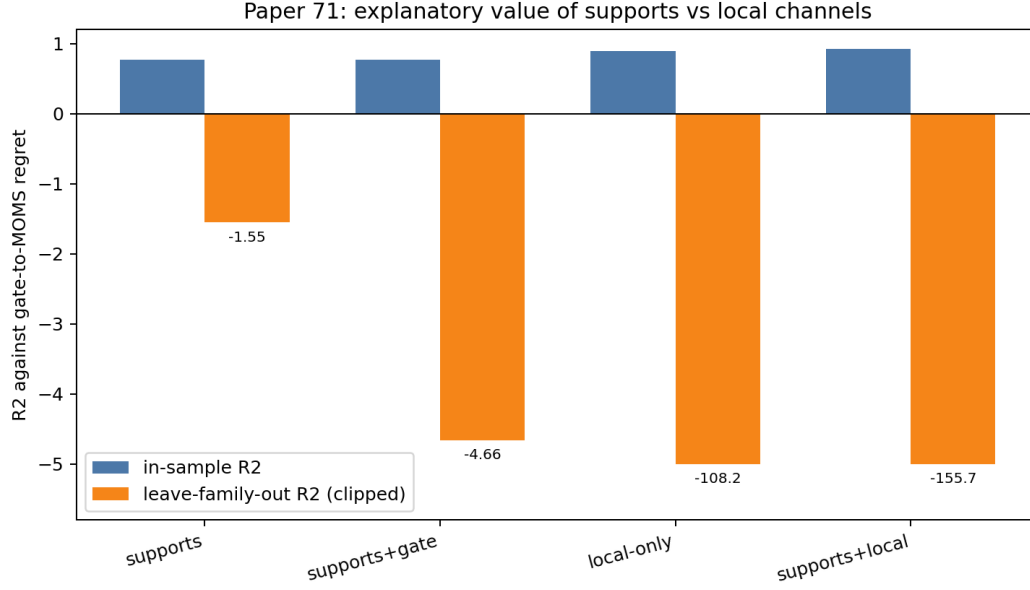


Figure 2: Support and local-channel models for gate-to-MOMS regret. Local channels improve in-sample fit but fail catastrophically under leave-one-family-out transfer. This is the central reason Paper 71 remains a hypothesis-generation paper.

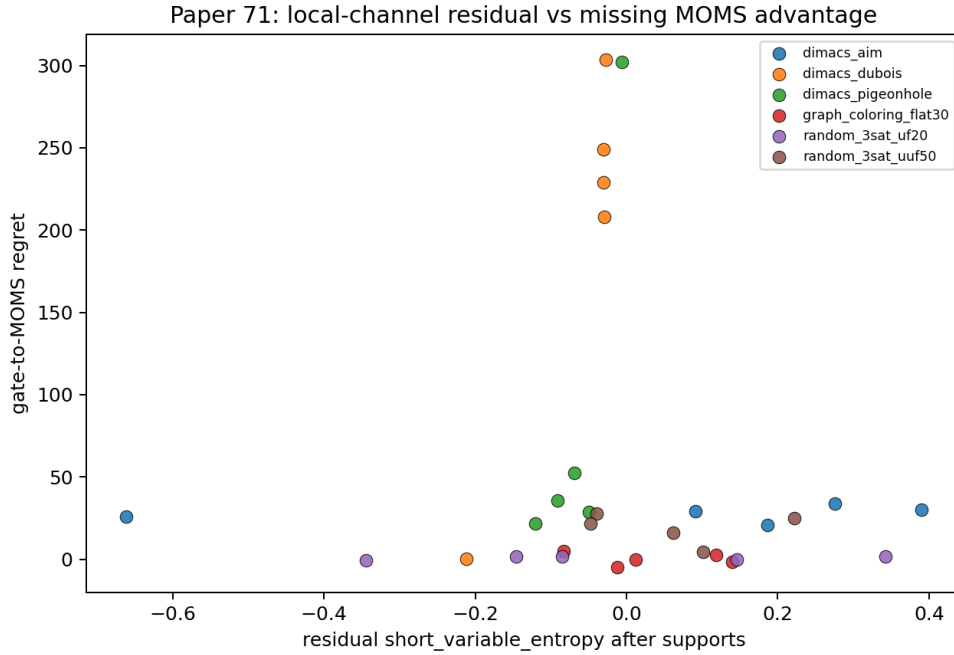


Figure 3: Residual shortest-clause variable entropy after support reconstruction. The strongest raw Paper-70 channel does not retain a residual relationship with gate-to-MOMS regret.

5 Interpretation

Paper 71 narrows the Paper-70 hypothesis. The missing MOMS channel is not simply:

“add short-clause entropy.”

The strongest raw channel from Paper 70 is largely absorbed by the existing support language. Likewise, raw short-clause pressure and MOMS-local pressure are mostly support-reconstructible.

The remaining candidate is subtler:

decision-local occurrence/degree residual structure.

This residual is weak, small-sample, and hypothesis-generating only. It should be read as a candidate for future falsification, not as evidence that a new SAT channel has already been established. It suggests that a future SAT-specific hypothesis should not add a broad “local structure” term. It should target the part of occurrence geometry not already represented by $(H, B, S, V, R_{\text{rob}}, G_{\text{gate}})$.

This is a useful outcome. Paper 71 prevents premature theory expansion. It says:

There is not yet enough evidence for a new primitive support or a MAAT v1.8. There is enough evidence to formulate a narrower SAT-specific local-occurrence hypothesis for a future preregistered experiment.

6 Limitations

The limitations are central to the interpretation:

- The analysis is post-hoc.
- The sample contains only 30 SATLIB smoke instances.
- The family structure is strong.
- Leave-one-family-out performance is poor for local-heavy models.
- Residual signals are weak.
- No solver policy is changed.
- No new gate is validated.

The correct output of this paper is therefore a candidate hypothesis, not an algorithm.

7 Reproducibility and Data Policy

Repository:

<https://github.com/Chris4081/structural-selection-principle>

Experiment folder:

experiments/sat_local_channel_independence_paper71/

Run command:

python3 analyze_local_channel_independence_paper71.py

Paper 71 uses only derived Paper-70 CSV outputs. No raw SATLIB CNF files are required or redistributed. The output summary explicitly marks the analysis as:

post-hoc-hypothesis-generation-not-gate-update.

8 Conclusion

Paper 71 tests whether the local SAT channel suggested by Paper 70 is independent of the existing MAAT support language. The answer is mixed and deliberately conservative. Most local channels are already strongly encoded by the support basis. The strongest raw Paper-70 signal disappears after support reconstruction. Only weak occurrence/degree residuals remain as candidates.

Therefore Paper 71 does not justify a new gate, a new MAAT version, or a retroactive reinterpretation of Paper 69. It justifies one next step: a future preregistered SAT-specific experiment testing whether decision-local occurrence residuals improve branching utility against MOMS and Jeroslow–Wang under the Gate Challenge discipline.

References

- [1] H. H. Hoos and T. Stützle, SATLIB: An Online Resource for Research on SAT, IOS Press, 2000. Available at <https://www.cs.ubc.ca/~hoos/SATLIB/>.
- [2] R. G. Jeroslow and J. Wang, “Solving propositional satisfiability problems,” *Annals of Mathematics and Artificial Intelligence* **1**, 167–187 (1990).
- [3] D. Pretolani, “Efficiency and stability of hypergraph SAT algorithms,” in *Cliques, Coloring, and Satisfiability: Second DIMACS Implementation Challenge*, DIMACS Series in Discrete Mathematics and Theoretical Computer Science, Vol. 26, American Mathematical Society (1996).
- [4] C. Krieg, “The Gate Challenge: Predeclared Falsification Protocol for MAAT v1.7,” MAAT Structural Selection Series, Paper 68 (2026). Zenodo DOI: <https://doi.org/10.5281/zenodo.20882852>.
- [5] C. Krieg, “Gate Challenge I: SATLIB/CDCL Smoke Execution,” MAAT Structural Selection Series, Paper 69 (2026).
- [6] C. Krieg, “Local Structural Channels in SAT,” MAAT Structural Selection Series, Paper 70 (2026).